Items to rate (36 MRs) – readable form

Each MR runs the program twice. JIR = how the two *inputs* relate; JOR = the *output* relation that must hold. Subscripts $1, 2$ = first/second call; `re/im` = real/imag part; `this/other` = operands; `out` = return. Classify each into one MR family (a–j) or `orphan` (see CODEBOOK).

ID	subject / mr_name	JIR / JOR
M01	ComplexSignal.add.0 / G_swap_negate	$(\text{this.re}_1 \approx \text{other.re}_2) \wedge (\text{this.im}_1 \approx \text{other.im}_2) \wedge (\text{other.re}_1 \approx \text{this.re}_2) \wedge (\text{other.im}_1 \approx \text{this.im}_2)$ $\Rightarrow (\text{out.re}_1 \approx \text{out.re}_2) \wedge (\text{out.im}_1 \approx \text{out.im}_2)$
M02	ComplexSignal.add.0 / L_scale	$(\text{this.re}_1 \approx 2.0 \cdot \text{this.re}_2) \wedge (\text{this.im}_1 \approx 2.0 \cdot \text{this.im}_2) \wedge (\text{other.re}_1 \approx 2.0 \cdot \text{other.re}_2) \wedge (\text{other.im}_1 \approx 2.0 \cdot \text{other.im}_2)$ $\Rightarrow (\text{out.re}_1 \approx 2.0 \cdot \text{out.re}_2) \wedge (\text{out.im}_1 \approx 2.0 \cdot \text{out.im}_2)$
M03	ComplexSignal.add.0 / T_shift_re	$((\text{this.re}_1 \approx (\text{this.re}_2 + 1.0))) \wedge (\text{this.im}_1 \approx \text{this.im}_2) \wedge (\text{other.re}_1 \approx \text{other.re}_2) \wedge (\text{other.im}_1 \approx \text{other.im}_2)$ $\Rightarrow ((\text{out.re}_1 \approx (\text{out.re}_2 + 1.0))) \wedge (\text{out.im}_1 \approx \text{out.im}_2)$
M04	MathSignalClass.clamp.0 / G_negate_bound_swap	$((lo_1 \approx (-hi_2))) \wedge ((hi_1 \approx (-lo_2))) \wedge ((x_1 \approx (-x_2)))$ $\Rightarrow \text{out}_1 \approx (-\text{out}_2)$
M05	MathSignalClass.clamp.0 / L_scale	$(lo_1 \approx 2.0 \cdot lo_2) \wedge (hi_1 \approx 2.0 \cdot hi_2) \wedge (x_1 \approx 2.0 \cdot x_2)$ $\Rightarrow \text{out}_1 \approx 2.0 \cdot \text{out}_2$
M06	MathSignalClass.clamp.0 / O_le_mono_in_x	$(lo_2 \approx lo_1) \wedge (hi_2 \approx hi_1) \wedge (x_2 \le x_1)$ $\Rightarrow \text{out}_2 \le \text{out}_1$
M07	MathSignalClass.clamp.0 / T_shift	$((lo_1 \approx (lo_2 + 1.0))) \wedge ((hi_1 \approx (hi_2 + 1.0))) \wedge ((x_1 \approx (x_2 + 1.0)))$ $\Rightarrow \text{out}_1 \approx (\text{out}_2 + 1.0)$
M08	MathSignalClass.exactLog2.0 / L_idem_at_one	$n_1 = 2 \cdot n_2$ $\Rightarrow \text{out}_1 = (\text{out}_2 + 1)$
M09	MathSignalClass.exactLog2.0 / O_le_monotone	$n_2 > 0 \wedge (n_2$
$(n_2 - 1)) = 0 \wedge n_1 > 0 \wedge (n_1$	$(n_1 - 1)) = 0 \wedge n_2 \le n_1$ $\Rightarrow \text{out}_2 \le \text{out}_1$	
M10	MathSignalClass.exactLog2.0 / T_double	$n_1 \approx 2.0 \cdot n_2$ $\Rightarrow \text{out}_1 \approx (\text{out}_2 + 1.0)$
M11	MathSignalClass.gcdSig.0 / G_swap_negate	$(a_1 = b_2) \wedge (b_1 = a_2)$ $\Rightarrow \text{out}_1 = \text{out}_2$
M12	MathSignalClass.gcdSig.0 / L_scale	$a_1 = 2L \cdot a_2 \wedge b_1 = 2L \cdot b_2$ $\Rightarrow \text{out}_1 = 2L \cdot \text{out}_2$
M13	MathSignalClass.hypotSig.0 / G_swap_negate	$(a_1 \approx b_2) \wedge (b_1 \approx a_2)$ $\Rightarrow \text{out}_1 \approx \text{out}_2$
M14	MathSignalClass.hypotSig.0 / L_scale	$(a_1 \approx 2.0 \cdot a_2) \wedge (b_1 \approx 2.0 \cdot b_2)$ $\Rightarrow \text{out}_1 \approx 2.0 \cdot \text{out}_2$
M15	MathSignalClass.hypotSig.0 / O_le_mono_in_abs	$(\lvert a_2 \rvert \le \lvert a_1 \rvert) \wedge (b_2 \approx b_1)$ $\Rightarrow \text{out}_2 \le \text{out}_1$
M16	MathSignalClass.isSequence.0 / B_rel_xor_reverse	$(a_1 \approx c_2) \wedge (b_1 \approx b_2) \wedge (c_1 \approx a_2)$ $\Rightarrow !(\text{out}_2 \wedge \text{out}_1)$
M17	MathSignalClass.isSequence.0 / G_negate_reverse	$((a_1 \approx (-c_2))) \wedge ((b_1 \approx (-b_2))) \wedge ((c_1 \approx (-a_2)))$ $\Rightarrow \text{out}_1 = \text{out}_2$
M18	MathSignalClass.isSequence.0 / L_scale_pos	$(a_1 \approx 2.0 \cdot a_2) \wedge (b_1 \approx 2.0 \cdot b_2) \wedge (c_1 \approx 2.0 \cdot c_2)$ $\Rightarrow \text{out}_1 = \text{out}_2$
M19	MathSignalClass.isSequence.0 / O_le_transitivity	$(a_1 \le a_2) \wedge (b_2 \le b_1) \wedge (c_2 \le c_1)$ $\Rightarrow (!\text{out}_2) \vee \text{out}_1$

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M20	MathSignalClass.isSequence.0 / T_rev_exclusion	$(a_1 \approx c_2) \wedge (b_1 \approx b_2) \wedge (c_1 \approx a_2)$ $\Rightarrow !(\mathrm{out}_2 \wedge \mathrm{out}_1)$
M21	MathSignalClass.isSequence.0 / T_shift	$((\lvert(a_1 - a_2) - (b_1 - b_2)\rvert < \varepsilon)) \wedge ((\lvert(b_1 - b_2) - (c_1 - c_2)\rvert < \varepsilon))$ $\Rightarrow \mathrm{out}_1 = \mathrm{out}_2$
M22	MathSignalClass.lcmSig.0 / G_swap_negate	$(a_1 = b_2) \wedge (b_1 = a_2)$ $\Rightarrow \mathrm{out}_1 = \mathrm{out}_2$
M23	MathSignalClass.lcmSig.0 / L_scale	$(a_1 = 2 \cdot a_2) \wedge (b_1 = 2 \cdot b_2)$ $\Rightarrow \mathrm{out}_1 = 2 \cdot \mathrm{out}_2$
M24	MathSignalClass.midpoint.0 / G_swap_negate	$(a_1 \approx b_2) \wedge (b_1 \approx a_2)$ $\Rightarrow \mathrm{out}_1 \approx \mathrm{out}_2$
M25	MathSignalClass.midpoint.0 / L_scale	$(a_1 \approx 2.0 \cdot a_2) \wedge (b_1 \approx 2.0 \cdot b_2)$ $\Rightarrow \mathrm{out}_1 \approx 2.0 \cdot \mathrm{out}_2$
M26	MathSignalClass.midpoint.0 / O_le_mono_in_a	$a_2 \leq a_1 \wedge (b_2 \approx b_1)$ $\Rightarrow \mathrm{out}_2 \leq \mathrm{out}_1$
M27	MathSignalClass.midpoint.0 / T_shift	$((a_1 \approx (a_2 + 1.0))) \wedge ((b_1 \approx (b_2 + 1.0)))$ $\Rightarrow \mathrm{out}_1 \approx (\mathrm{out}_2 + 1.0)$
M28	MathSignalClass.powerSig.0 / D_reciprocal_exp	$(base_1 \approx base_2) \wedge base_2 \neq 0.0 \wedge n_1 = -n_2$ $\Rightarrow \mathrm{out}_1 \approx (1.0/\mathrm{out}_2)$
M29	MathSignalClass.powerSig.0 / E_power_of_power	$base_1 \approx \mathrm{out}_2$ $\Rightarrow \mathrm{out}_1 \approx powerSig(base_2, n_2 \cdot n_1)$
M30	MathSignalClass.powerSig.0 / G_odd_negate	$(n_1 = n_2) \wedge ((base_1 \approx (-base_2)))$ $\Rightarrow ((n_2 M31$
MathSignalClass.powerSig.0 / L_scale_base	$(base_1 \approx 3.0 \cdot base_2) \wedge (n_1 = n_2)$ $\Rightarrow \mathrm{out}_1 \approx (powerSig(3.0, n_2) \cdot \mathrm{out}_2)$	
M32	MathSignalClass.powerSig.0 / T_exp_step	$base_1 = base_2 \wedge (n_1 = (n_2 + 1))$ $\Rightarrow \mathrm{out}_1 \approx (\mathrm{out}_2 \cdot base_2)$
M33	MathSignalClass.signum.0 / D_reciprocal	$x_2 \neq 0.0 \wedge (x_1 \approx (1.0/x_2))$ $\Rightarrow \mathrm{out}_1 = \mathrm{out}_2$
M34	MathSignalClass.signum.0 / G_negate	$x_1 \approx (-x_2)$ $\Rightarrow \mathrm{out}_1 = -\mathrm{out}_2$
M35	MathSignalClass.signum.0 / L_scale_pos	$x_1 \approx 3.0 \cdot x_2$ $\Rightarrow \mathrm{out}_1 = \mathrm{out}_2$
M36	MathSignalClass.signum.0 / O_le_monotone	$x_2 \leq x_1$ $\Rightarrow \mathrm{out}_2 \leq \mathrm{out}_1$